



CONTAINER LIFECYCLE MANAGEMENT SYSTEM

Higher Diploma in Science in Computing

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1. Introduction

Background

In pharmaceutical manufacturing, product containers are reused throughout production and must be carefully managed to ensure product quality, traceability, and compliance with Good Manufacturing Practice (GMP) requirements. Before a new container can be introduced into production, it must undergo cleaning and receive Quality Assurance (QA) approval to confirm it is fit for use.

Once approved, the container enters its production lifecycle, allowing it to be used for a maximum of 14 production runs or 30 calendar days, whichever comes first. When the limit is reached, the container must be removed from production and sent for cleaning. Following cleaning, the container must be approved by a supervisor before being reintroduced into the production area, at which point a new lifecycle begins.

Problem Statement

Container management has traditionally relied on paper-based logbooks and manual documentation, making the process prone to human error. This project proposes a Container Management System that digitises the lifecycle of each container by recording movements, approvals, and usage. The system enforces lifecycle rules, prevents expired containers from being used in production, reduces the risk of costly investigations and delayed batch release, and improves traceability, accuracy, and compliance with GMP requirements.

2. Project Scope and Objectives

The aim of this project is to develop a full-stack web application that manages the complete lifecycle of reusable production containers used in pharmaceutical manufacturing. The system will track each container from its initial introduction through cleaning, Quality Assurance (QA) approval, production use, and subsequent cleaning cycles. It will provide end-to-end traceability while enforcing business rules that ensure containers remain compliant with Good Manufacturing Practice (GMP) requirements.

The project scope includes:

- User Authentication
- Role-based access control
- Container lifecycle management
- Workflow validation
- QA approval workflows
- Supervisor approval workflows
- Observation management
- Audit logging
- Operational dashboard
- Predictive analytics

The objectives of the project are to:

- Design a relational MySQL database.
- Develop a RESTful API using Node.js and Express.
- Develop a React-based frontend.
- Implement workflow validation.
- Implement role-based approval process.
- Record observations and audit logs.
- Generate simulated operational data for testing and analysis.
- Develop analytics dashboards to visualise operational performance.
- Apply regression analysis to predict container turnaround times.

3. Student's Learning Objectives

The learning objectives for this project are to develop and apply software engineering knowledge while building a real-world solution to a pharmaceutical manufacturing problem. The project aims to strengthen full stack development skills, gain experience with modern web technologies, and explore data-driven techniques that support operational decision-making.

The learning objectives are to:

- Improve full-stack development skills by learning modern React development.
- Apply software engineering principles throughout the software development lifecycle.
- Develop a secure RESTful web application using Node.js, Express, React, and MySQL.
- Generate realistic datasets through simulation for testing and analysis.
- Explore predictive analytics by applying regression techniques to operational data.
- Gain practical experience in designing software solutions for pharmaceutical manufacturing.

4. Technical Specification of the Project

This project was developed using a modern full-stack technology stack to support the design, implementation, testing, and deployment of the Container Management System. The selected hardware and software tools were chosen to provide a reliable development environment while supporting secure application development, database management, API testing, version control, and data visualisation. **Hardware Used**

- Windows PC
- Internet connection

Software Used

- Visual Studio Code (Code editor)
- Node.js (JavaScript runtime environment)
- Express.js (Backend web framework)
- React (Frontend JavaScript library)
- MySQL (Relational database management system)
- MySQL Workbench (Database design and administration)
- Git (Version control system)
- GitHub (Source code repository)
- Postman (API development and testing)
- Chart.js (Data visualisation library)
- npm (Package manager)
- Jest (JavaScript testing framework)

Database Used

- **MySQL**

Version Control

- Git
- GitHub

5. Project Plan

The project followed a ten-week development plan using an incremental approach. Each week focused on a specific stage of the software development lifecycle, allowing requirements, design, implementation, testing, and documentation to be completed in a structured and manageable manner. This approach ensured that each component of the application was developed and tested before progressing to the next stage.

Week 1

- Requirements gathering
- Research
- Entity Relationship Diagram (ERD)

Week 2

- Database implementation
- MySQL schema

Week 3

- Backend API
- API testing using Postman
- Unit testing using Jest

Week 4

- Authentication
- Workflow engine
- Authentication and role-based testing

Week 5

- Operational Data Generator

Week 6

- React frontend

Week 7

- Dashboard implementation

Week 8

- Regression analysis
- Analytics testing

Week 9

- System testing
- Bug Fixing

Week 10

- Documentation
- Final report

6. Conclusion

The project addresses a real-world operational problem by providing a digital solution for managing the lifecycle of reusable production containers. The system combines workflow management with predictive analytics to improve traceability, support better decision-making, and ensure compliance with Good Manufacturing Practice (GMP) requirements. The completed application demonstrates modern software engineering principles while providing valuable insights through predictive modelling.

The system complements existing GMP documentation by becoming part of the controlled operational process. Integrating the application with standard operating procedures and existing paperwork encourages consistent use of the digital system, improving traceability, data integrity, and regulatory compliance.

Ultimately, the system supports the shared objective of producing safe, high-quality pharmaceutical products by improving process control, traceability, and compliance throughout the container lifecycle. By reducing the likelihood of expired or non-compliant containers entering production, the system also has the potential to minimise costly investigations, reduce batch release delays, and improve overall operational efficiency.

7. References / Bibliography

React is an open-source JavaScript library used for building interactive user interfaces through reusable components. Its component-based architecture promotes modular software development making applications easier to maintain and extend. Research has shown that React's efficient rendering model and Virtual DOM make it well suited to modern web applications that require responsive user interfaces. These characteristics make React an appropriate choice for the Container Management System. Where dynamic dashboard, container status update, and approval workflows require a responsive and maintainable frontend.

(React, 2026) (Hoque, 2020) (Devanshi Bhatt, 2023)

Node.js was selected as the server-side runtime environment because its asynchronous, event-driven architecture enables efficient handling of multiple client requests. This makes it well suited to web applications that require frequent communication between the frontend, backend, and database, such as the Container Management System.

(Ghansham Jadhav1, 2020) (Hoque, 2020) (Node.js, 2026)

Express.js was selected because it provides a lightweight and flexible framework for developing RESTful APIs. It's routing and middleware capabilities support the implementation of authentication, workflow validation, and approval processes required by the Container Management System.

(Hoque, 2020) (Express.js, 2026) (MDN Web Docs, 2026)

MySQL was selected because its relational database model supports structured data storage and maintains integrity through primary and foreign key relationships. These features are well suited to the Container Management System, where container movements, approvals, observations, and audit records are linked across multiple database tables.

(Williams, 2007) (MySQL, 2026) (Annala, 2026)

Chart.js was selected because it provides an efficient and lightweight solution for creating charts within the React-based dashboard. This enables operational data, such as container usage, turnaround times, and lifecycle trends, to be presented in a clear and meaningful format and support informed decision-making.

(Chart.js, 2026) (Cairo, 2019) (Samarasinghe, 2026)

Workflow management involves analysing existing business processes (the Asis process) to understand current activities and identify inefficiencies.

Organisations then design a To-be process that improves efficiency, consistency, and control through redesigned workflows and automation. In this project, the existing paper-based container management process was analysed before designing a digital workflow. This new system will automate approvals, enforce lifecycle rules, and improve traceability throughout the container's lifecycle.

(by Alec Sharp, 2008) (Villanueva, 2026)

Traceability is a fundamental requirement within pharmaceutical manufacturing, enabling organisations to monitor the movement and status of materials throughout the manufacturing process. Effective traceability supports regulatory compliance, facilitates investigations, and helps ensure product quality and patient safety. The proposed Container Management System enhances traceability by recording container movements, approvals, observations, and lifecycle events with a centralised digital audit trail.

(Ducellier, 2016) (MANAPHARMA, 2026) (Agency, 2026)

Predictive analytics applies statistical techniques and historical data to forecast future events and support organisational decision-making. According to Siegel, every predictive analytics application consists of two key elements: determining what behaviour or event is to be predicted and deciding what action should be taken based on that prediction. Within the Container Management System, predictive analytics is applied to estimate container turnaround times using regression analysis. These predictions provide operational insights that support planning, improves container availability, and reduces risk of production delays.

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